JINJA JOINT EXAMINATINS BOARD

JJEB P510/1 SCORING GUIDE 2026

ITEM 1

(a) $^{x}/_{25}$

mass of reactants = 235.1 + 1.009 = 236.109

mass of products = (148.0 + 84.9) + (3 × 1.009) = 235.927U√

Energy released = $mass\ defect \times 931MeV$ √

Energy released = $(236.109 - 235.927) \times 931MeV$

Energy released = $169.442MeV$√

Energy released = $169.442 \times 10^{6} \times 1.6 \times 10^{-19} = 2.71 \times 10^{-11}J$√

This energy is released by one atom

253g of Uranium-235 contains 6.02×10^{23} atoms

1kg contains $\frac{6.02\times10^{23}}{235\times10^{-3}} = 2.562 \times 10^{24}$ atoms√

Energy released by One atom = $2.71 \times 10^{-11}J$

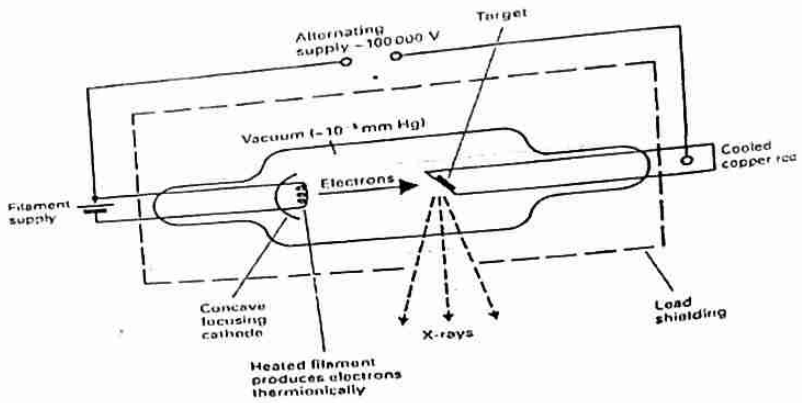
2.562×10^{24} atoms release $2.562 \times 10^{24} \times 2.71 \times 10^{-11}$√ $= 6.943 \times 10^{13}J$√

Energy released by 1kg of uranium = $6.943 \times 10^{13}J$

The energy released is $6.943 \times 10^{13}J$ which is greater than 6.0 x 10^{13}J the target energy. √

8 scores

(b) The device used to detect the broken bones is an X-ray tube



any 4 labelled parts

✓✓✓✓

Operation

- The cathode is heated with low voltage and electrons are emitted thermionically. ✓
- Electrons are accelerated by a high p.d towards the anode. ✓
- On striking the target, about 99% of their kinetic energy is converted into heat and 1% converted to X-rays ✓

7 scores

How to detect bone fracture

The soft X-rays✓ are used and are directed towards a suspected part of the body.

A photographic plate is placed behind the body✓.

The X-rays are absorbed by the bones while those that pass through the muscle are able to reach the plate and darken it. ✓

The white part on the plate shows the bone and dark lines in the white part show the cracks within the bone. ✓

4 scores

Dangers of radiations

- They cause genetic mutation ✓
- They cause skin burn✓
- They damage body cells✓

- They cause sterility
- They damage eye sight

3 scores

Precautions

- Lead aprons should be worn when dealing with radiations ✓
- Avoid unnecessary exposure to the radiations ✓
- Delicate parts should not be exposed to the radiations ✓

3 scores

ITEM 2

(a) Surfaces rest on each other on small cross sectional areas✓, thus the actual area of contact between the two surfaces is very small✓, hence the pressure at points of contact is very high✓. Projections merge to produce welding before the motion can take place these welds have to be broken✓ hence the force which opposes moti is obtained✓

5 scores

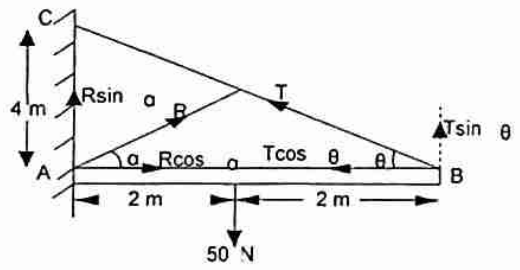
Forces acting on the beam are

Weight of the beam✓

Resolved components of the tension✓

Resolved components of the Normal reaction✓

3 scores



✓

$\tan\theta = \frac{4}{4}$ $\therefore\theta = 45°$ ✓

Taking moments about A

$Tsin\,\theta x4 = 50x2$ ✓

$Tsin45x4 = 50x2$

$T = 35.36N$

Resolving vertically: $Rsin\,\alpha + Tsin\,\theta = 50$

$Rsin\,\alpha + 35.36sin45 = 50$

$Rsin\,\alpha = 24.997$✓.....(i)

Resolving horizontally $Rcos\alpha = Tcos\theta$

$Rcos\alpha = 35.36cos45$

$R\cos\alpha = 25$ .....(ii) ✓

(i)/(ii)

$$\frac{R\sin\alpha}{R\cos\alpha} = \frac{24.997}{25}$$

$\tan\alpha = \frac{24.997}{25} \therefore \alpha = 45^0$ ✓

8 scores

Put into (ii)

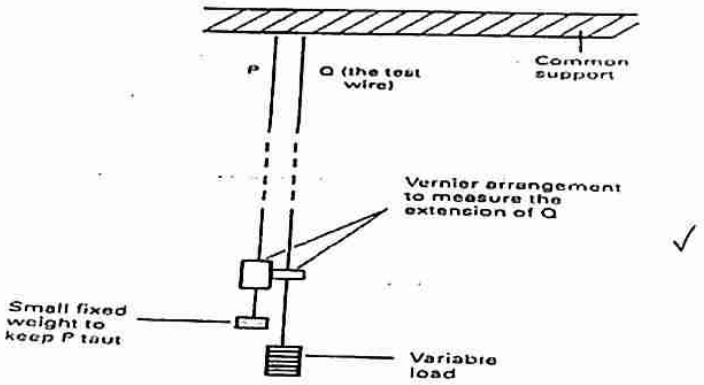
$R\cos\alpha = 25$

$R\cos 45 = 25$

R = 35.36N at 45^0 to the beam✓

Since the normal reaction at the hinge is less than 40N the it will last longer✓

(b)



✓

- A spirit level L indicates whether or not, it is horizontal.

Initial weights are placed at the end of P and Q to remove any kinks and the spirit ✓

Two identical steel wires P and Q that are suspended from the same rigid support are attached to two frames connected by a spirit level. ✓

level is put back to the horizontal level by turning micrometer screw gauge.

Additional masses m are added to free end of Q and the extension is indicated by the micrometer scale.

The procedure is repeated for various masses m. however micrometer readings are also taken when the loads m are offloaded. This is to verify whether elastic limit has not reached. ✓

The diameter d of the wire is measured using the micrometer screw gauge at several places and their average calculated. The cross-sectional area A is calculated and recorded $A = \frac{\pi d^2}{4}$ ✓

The original length l of wire Q is measured using a metre rule. ✓

The extension x is measured for different values of mass m. ✓

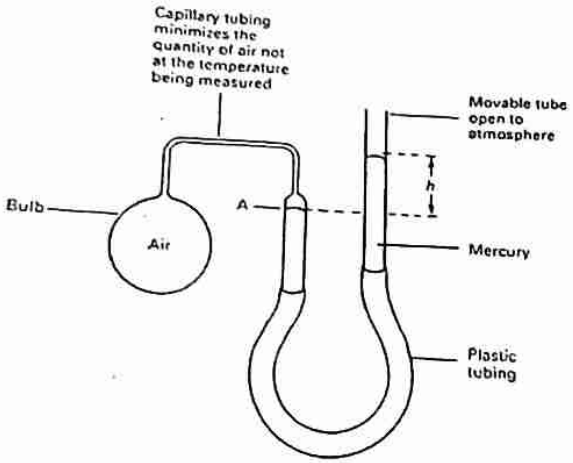
A graph of load against extension is plotted and its slope S obtained✓

The young's modulus of the material is got from $E = \frac{4Sl}{\pi d^2}$ ✓

10 scores

ITEM 3

(a) The device is constant volume thermometer✓

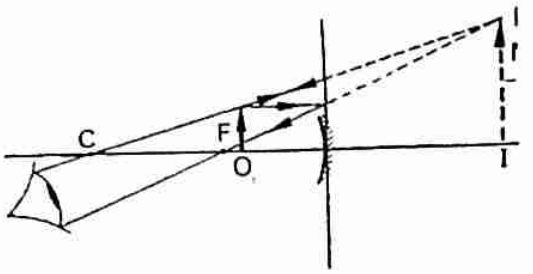


✓ ✓ any two correct labels

- The bulb with air is placed in a room whose temperature is required. ✓
- The bulb is given enough time so at the gas acquires the temperature of the room, the gas warms up and expands pushing the mercury level in the closed limb and up in the open limb✓
- By adjusting the plastic tubing up and down, the level in A is restored keeping the volume constant. ✓
- The difference in mercury levels h is determined and the thermometer reading H due to atmosphere in the open limb is recorded ✓
- The total pressure, P_θ exerted by the gas at temperature,θ is obtained from $P_\theta = H + h$. ✓
- The pressure is then measured at the ice point P_0, at steam point P_{100}, by the same procedure
- Therefore the temperature, θ on this thermometer is obtained from $\theta = (\frac{P_\theta - P_0}{P_{100} - P_0}) x100°C$ ✓

09 scores

Addressing the concern f the patient



✓ object between f and P

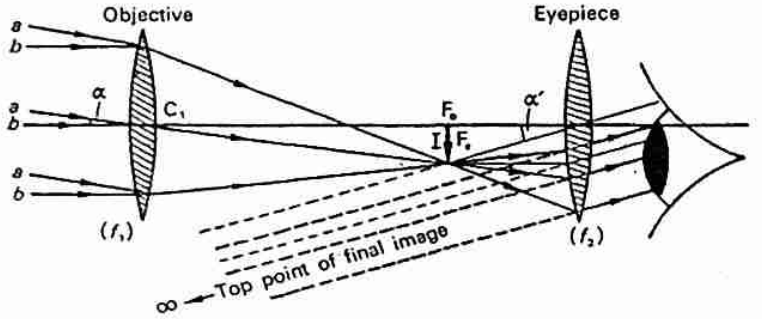
✓ correct rays

✓ virtual image

The tooth is placed between the F and the pole✓. The concave mirror forms a magnified, Erect and virtual image of the tooth. ✓

05 scores

(b)



✓ objective

✓eyepiece

✓correct rays

The image of the distant object is formed by the objective lens at its principal focus and the image formed is real and inverted✓. This image acts as the object of the eyepiece and it is adjusted to that the object lies on the principal focus✓ of eyepiece and forms the final image at infinity which virtual and magnified✓

Aangular magnification, $m = \frac{\alpha^1}{\alpha}$✓

For small angles in radians: $\alpha^1 \approx tan\alpha^1 = \frac{h_1}{f_e}$

$$\alpha \approx tan\alpha = \frac{h_1}{f_o}$$

$$m = \frac{\alpha^1}{\alpha} = \frac{\left(\frac{h_1}{f_E}\right)}{\left(\frac{h_1}{f_o}\right)}\checkmark = \frac{f_o}{f_e}$$

$m = \frac{f_o}{f_e}$✓

$m = \frac{300}{25} = 12$✓

11 scores

Since the angular magnification is greater than 6 then it's a good instrument for viewing a distant object. ✓

ITEM 4

(a) Formation of stationary waves

A stationary wave is formed as a result of superposition ✓of two progressive waves of equal amplitude✓ and frequency✓ but travelling at same speed in opposite direction. ✓

Stationary waves are characterized by node (N) and antinodes (A) ✓

05 scores

Determination of frequency of a note from guitar

- A note is sounded together with a tuning fork of known frequency, f_T✓
- The number of beats, n in t seconds are counted✓ and the beat frequency, $f_b = \frac{n}{t}$ calculated✓
- One prong of the tuning fork is loaded with plasticine and then the experiment repeated✓. The new beat frequency $f_b{}^1$ is determined✓
- If $f_b{}^1 > f_b$ then the frequency of the test note f_n will be the lower value and is calculated from $f_n = f_T - f_b$✓
- If $f_b{}^1 < f_b$ then the frequency of the test note f_n will be the upper value and is calculated from $f_n = f_T + f_b$✓

07 scores

To determine the tension

$v = 2lf_o$✓ $= 2 \times 0.80 \times 200 = 320ms^{-1}$✓

$V = \sqrt{\left(\frac{T}{\mu}\right)}$ ✓

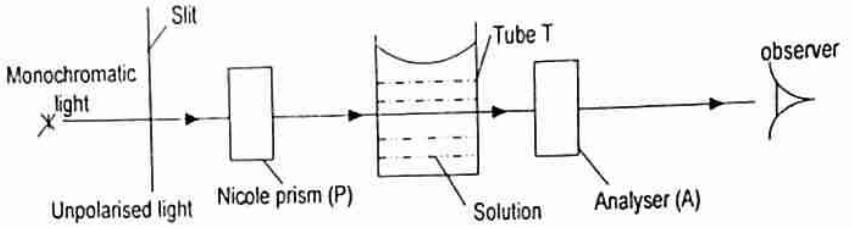
$320 = \sqrt{\left(\frac{T}{2.0\times10^{-3}}\right)}$ ✓

$T = 2.0 \times 10^{-3} \times 320^2 = 204.8\ N$✓

06 scores

Hence the guitar strings are well tuned✓

(c) To Measurement of concentration of sugar in solution



✓✓

Apparatus is arranged as above.

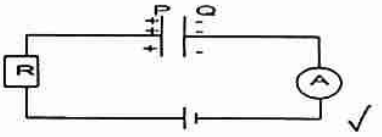
- Without tube T in place, the analyzer A is rotated until the emergent light from T is completely cut off. The position of A is noted. ✓
- Tube T is filled with a solution to be tested and on looking through A, light can now be seen✓
- Viewing through the analyzer A, it is rotated until when light is cut off and note this point. ✓
- Measure the angle of rotation θ of the analyzer. ✓
- The concentration of the solution is proportional to the angle of rotation therefore the concentration can be determined ✓

07 scores

ITEM 5

(a) The electrical device is a capacitor✓

Charging a capacitor



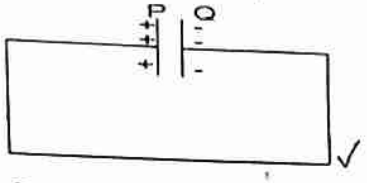
✓

When the capacitor is connected to a battery, electrons flow from the negative terminal of the battery to the adjacent plate of the capacitor ✓and at the same rate electrons flow from plate P of the capacitor towards the positive terminal of the battery leaving positive charges at P. ✓

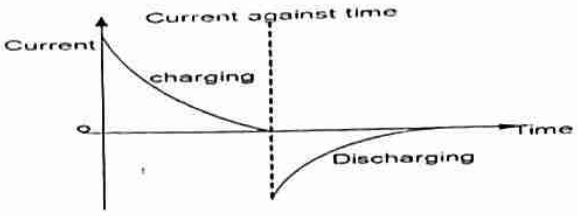
Positive and negative charges therefore appear on the plate and oppose the flow of electrons that cause them✓

As charge accumulates the p.d between the plates increase and charge current falls to zero when the p.d between the plates of the capacitor is equal to battery voltage✓

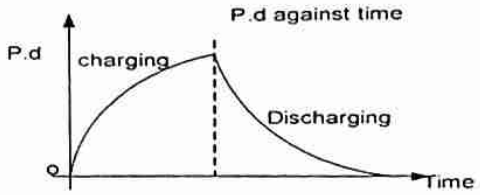
Discharging a capacitor



Connect a wire from the positive plate to the negative plate. Electrons flow from the negative plate to positive plate through neutralizing the positive charge✓. The electrons flow until all the charge has been neutralized and there is no charge on either plates of the capacitor the capacitor is fully discharged✓.



✓



✓

11 scores

(b) testing suitability of the heating coil

C = 200 μF, V = 12 V, 2 litres = 2000 cm³, using mass = density × volume✓

$Mass\ of\ water = 2000\ cm^3 \times 1\ gcm^{-3} = 2000\ g = 2.0\ kg$✓

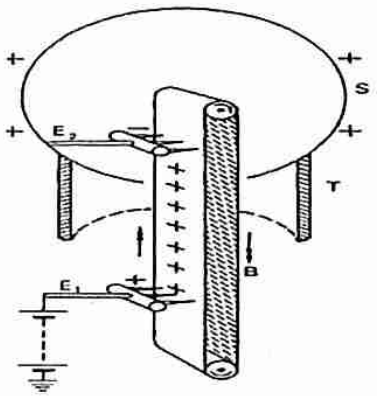
Using *Electrostatic potential energy = Heat gained by the water.* ✓

$\frac{1}{2}CV^2 = mc\Delta\theta \implies \frac{1}{2} \times 200 \times 10^{-6} \times 12^2 = 2.0 \times 4200 \times \Delta\theta$ ✓

$\therefore The\ temperature\ change, \Delta\theta = 1.71 \times 10^{-6}\ °C$ ✓

Hence the coil is meets the required specification✓

b) the device is the van de Graaff generator



✓✓✓ any 3 labelled parts

The electrode E_1 is made highly positive relative to the earth✓. The high electric field intensity at the sharp points of E_1 ionizes air around it repelling positive charges on to the belt. ✓

The belt driven by a motor carries this charge into the sphere. As it approaches E_2, it induces negative charge at the spikes of E_2 and positive charge on the outer surface of sphere✓

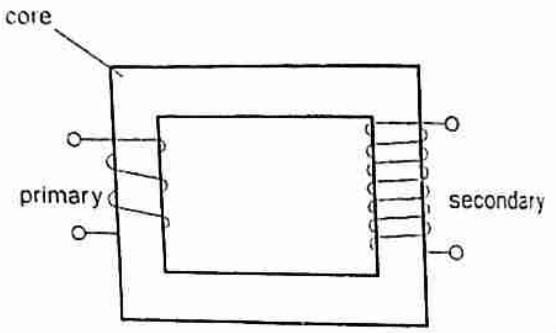
The high electric field intensity around E_2 ionizes air there, repelling negative charge onto the belt. The negative charge neutralizes positive charge on the belt before it goes over the upper pulley. ✓

The process is repeated many times until the potential of the sphere is about 10^6V relative to the earth ✓

14 scores

ITEM 6

(a)



✓✓ complete structure

Alternating voltage connected to the primary coil drives an alternating current in it. This sets up an alternating magnetic flux✓ in the core which links up the secondary coils and an emf is induced✓ in the secondary coils. The magnitude of the e.m.f induced in the secondary coil, in accordance to faraday's law is $V_S = -N_S A\frac{dB}{dt}$✓

The changing magnetic flux in the primary induces a back e.m.f✓ in the primary coil. The back emf opposes the applied p.d V_P . so the applied p.d V_P must be equal the back emf to keep the flux changing.

Thus $V_P = -N_P A\frac{dB}{dt}$

Thus step up voltage can be calculated from $\frac{V_P}{V_S} = \frac{N_P}{N_S}$✓

07 scores

Energy losses and how they are minimized

- Some of the energy is lost as heat due to the resistance of the coils i.e $P = I^2R$✓. This loss is minimized by using thick copper wires of low resistance. ✓

- Eddy currents✓: currents circulating in the core due to the changing magnetic flux is dissipated heat by I^2R mechanism. This results into the heating of the core. This is minimized by using a laminated core made of thin strips or laminar separated from each other by a layer of insulating varnish. ✓
- Hysteresis loss✓: energy used to turn the magnetic domains of the core is dissipated as heat
  It can be minimized by using a soft magnetic material such as soft iron. ✓
- Some loss of energy occurs because a small amount of flux associated with the primary coil may not pass through the secondary coil. This can be minimized by winding one coil on top of the other.

06 scores any 3

(b) Addressing concerns of mechanic

Maximum power output

Suppose the load resistance R is variable then the useful power expended in R will also vary and will be maximum when $R = r$.

$Power\ output = IV\checkmark = \left(\frac{E}{R+r}\right)x\left(\frac{ER}{R+r}\right)\checkmark$

$$Power\ output, P_0 = \frac{E^2}{(R+r)^2}R$$

At maximum power, P_m, $\frac{dP_0}{dR} = 0$

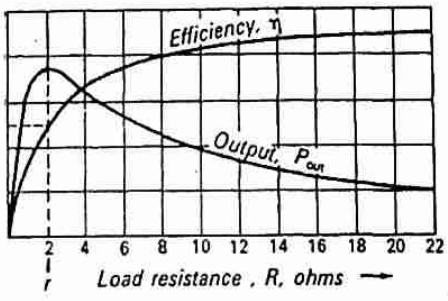
$\frac{dP_0}{dR} = E^2\frac{[(R+r)^2-2R(R+r)]}{(R+r)^4} = 0\checkmark$

$R = r\checkmark$

$P_0 = \frac{E^2}{(r+r)^2}r\checkmark$

Power output, $P_0 = \frac{E^2}{4r}\checkmark$

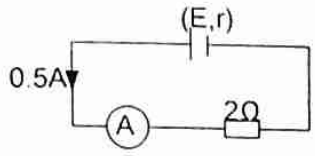
Variation of η and P with R



07 scores

✓efficiency ✓ power output

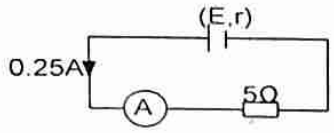
To calculate emf of the battery



$E = I\,(R + r)$

$E = 0.5\,(2 + r)$

$E = 1 + 0.5\,r$.....[1] ✓



$E = I\,(R + r)$

$E = 0.25\,(5 + r)$

$E = 1.25 + 0.25\,r$.....[2] ✓

Equating 1 and 2

$1 + 0.5\,r = 1.25 + 0.25\,r$

$0.5\,r - 0.25\,r = 1.25 - 1$

$0.25r = 0.25$

$r = \dfrac{0.25}{0.25}$

$r = 1\,\Omega$✓

Also $E = 1 + 0.5\,r$

$E = 1 + 0.5\,x\,1$

$E = 1 + 0.5$

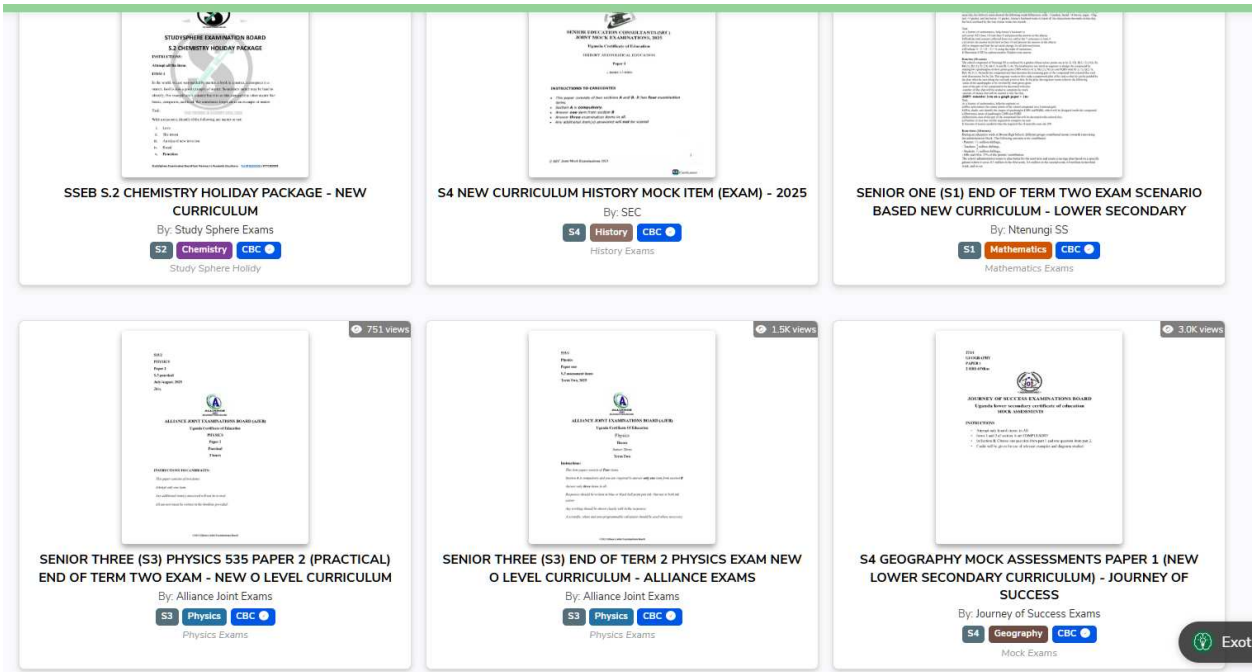
$E = 1.5\,V$✓

Hence the batter will start the car✓

05 scores

END

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